

Justification Documents

Purpose of the Script:

This Python script is intended to do an automated Windows System Health Check. The client wanted a routine that would be useful to a system administrator and did not specify which pywin32 modules to use. Based on this requirement I wrote a routine that gathers important information regarding the state of a Windows machine and combines it into one simple to read report.

In a real workplace, a system administrator often has to monitor many machines. Manually checking the computer name, server information, service status, running processes, and disk space on each computer is tedious and easy to forget. This program automates that process: it retrieves all five pieces of information in one run, outputs them to the screen, saves them to a text file for record-keeping, and displays a confirmation pop-up when it's finished. This reduces the amount of manual effort and lets the administrator catch problems, such as a stopped service or a nearly full disk, before they become bigger problems.

Module Justification:

- **Module 1: Computer name**

The win32api module is used to call GetComputerName(), which retrieves the name of the machine the script is running on. This is included first in the report because it identifies which computer the rest of the report belongs to. If an administrator is reviewing reports collected from multiple machines, this single line lets them immediately tell which machine each report came from.

- **Module 2: Operating System Information Check**

The win32net module is used to call NetServerGetInfo() at information level 101, which returns the platform ID and the OS version number (major and minor) of the local machine. This function supports the project's goal because it gives the administrator a quick way to confirm what platform and version the machine is running, which is useful when checking whether machines are up to date or compatible with certain software.

- **Module 3: Service Status Check**

The win32service module is used to open the Service Control Manager and query the status of a list of important Windows services: Spooler (printing), wuauclt (Windows Update), and WinDefend (Windows Defender). Rather than just calling a function once, the script loops through the list of services, checks the state of each one, and builds a list of any services that are stopped or missing. If any are

found, a warning line is added to the report. This function was chosen because these services directly affect a computer's usability and security.

For example, if Windows Defender is stopped, the machine is at risk, and this module would flag that immediately instead of requiring the administrator to check manually.

- **Module 4: Running Process Check**

The win32process module is used to call EnumProcesses() to get the IDs of every process currently running on the machine, and GetModuleFileNameEx() to identify each process by name. The script counts the total number of processes running and checks to see if three important processes are running: explorer.exe, svchost.exe, and winlogon.exe. We chose these processes because they are core to the Windows operating system: explorer.exe gives you the desktop and taskbar, winlogon.exe handles user login sessions, and svchost.exe houses a number of important background services. If any of these processes are not running, it is a strong indication that the system is badly broken, so this check gives the administrator early warning of major system problems.

- **Module 5: C: Disk Space Check**

This module can automatically flag low disk space so the administrator can fix the problem before it crashes the system. The win32file module is used to call GetDiskFreeSpaceEx() on the C: drive, which returns the free space and total size of the drive. The script calculates the percent free space remaining and if it's less than 15%, it places a low-disk-space warning in the report.

This is useful for the project since running out of disk space can cause serious problems like failed updates or system instability. This module can automatically flag low disk space so the administrator can fix the problem before it crashes the system.